St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question Bank (Revised Curriculum 2018 onwards)

First Year- Second Semester (2019 -21 Batch)

Course Title (Paper Title):

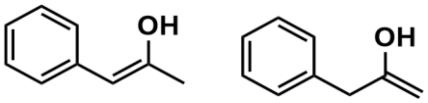
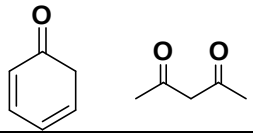
Synthetic Organic Chemistry and Molecular Rearrangements. QP Code:55103

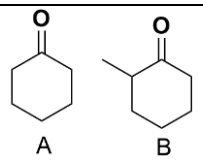
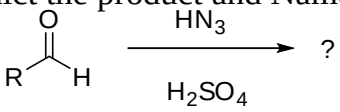
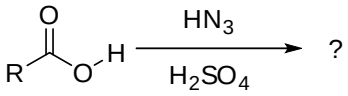
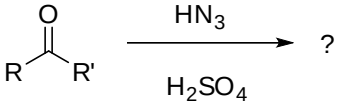
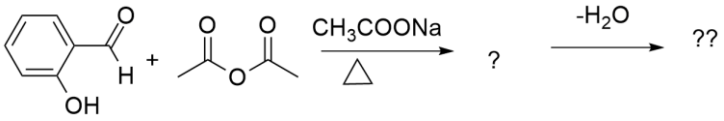
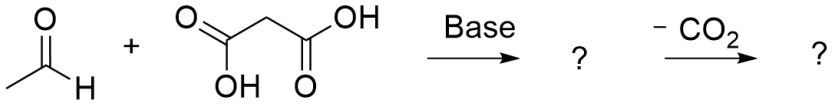
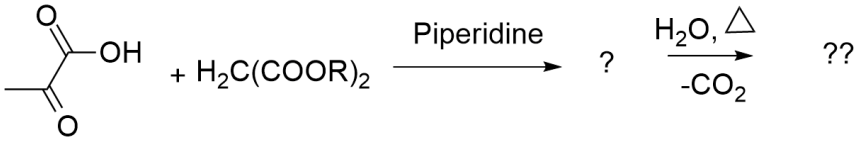
Unit	SI. No	Questions	Marks
1	1.	What is inter and intramolecular rearrangement?	2
1	2.	Give the types molecular rearrangement.	2
1	3.	Define inter and intramolecular rearrangement?	2
1	4.	What is Pinacol-Pinacolone rearrangement? Give an example.	2
1	5.	Name the reagent used and also the type of the product formed in the Pinacol-Pinacolone and Benzidine rearrangement.	2
1	6.	Give an example for Pinacol-Pinacolone rearrangement?	2
1	7.	Write an example for Benzidine rearrangement.	2
1	8.	Define Benzidine rearrangement? Give an example.	2
1	9.	What is Wagner-Meerwein rearrangement? Give its reaction.	2
1	10.	Mention the reagent used and also the type of the product formed in the Benzidine and Wagner-Meerwein rearrangement.	2
1	11.	Give an example for Demjanove rearrangement.	2
1	12.	What is Demjanove rearrangement? Give an example.	2
1	13.	Write an example for Benzilic acid rearrangement.	2
1	14.	What is Benzilic acid rearrangement? Give an example.	2
1	15.	Give an example for Favorskii rearrangement.	2
1	16.	Define Favorskii rearrangement? Give an example.	2
1	17.	Name the reagent used and also the type of the product formed in the Benzylic acid and Favorskii rearrangement.	2
1	18.	Give an example for Arnt-Eistert synthesis.	2
1	19.	Give Arnt-Eistert synthesis.	2

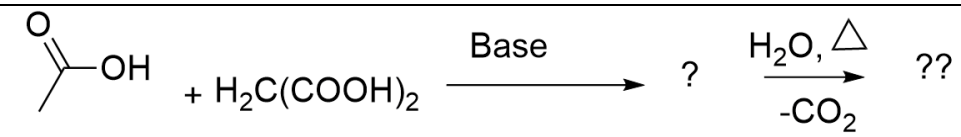
1	20.	Mention the reagent used and also the type of the product formed in the Arnt-Eistert synthesis and Fries rearrangement.	2
1	21.	What is Fries rearrangement? Give an example.	2
1	22.	Give an example for Fries rearrangement.	2
1	23.	What is Steven's rearrangement? Give an example.	2
1	24.	Give an example for Steven's rearrangement.	2
1	25.	Name the reagent used and also the type of the product formed in the Steven's and Dienophine rearrangement.	2
1	26.	What is Dienophine rearrangement? Write its reaction..	2
1	27.	Define Hofmann rearrangement? Give an example.	2
1	28.	Write an example for Hofmann rearrangement.	2
1	29.	Name the reagent used and also the type of the product formed in the Curtius and Schmidt rearrangement.	2
1	30.	What is Curtius rearrangement? Give an example.	2
1	31.	Give an example for Curtius rearrangement.	2
1	32.	What is Lossen rearrangement? Give an example.	2
1	33.	Illustrate the role of Lossen rearrangement.	2
1	34.	Give an example for Lossen rearrangement.	2
1	35.	Name the reagent used and the type of the product formed in the Lossen and Hofmann rearrangement.	2
1	36.	What is Schmidt rearrangement? Write its reaction.	2
1	37.	Give an example for Schmidt rearrangement.	2
1	38.	What is Beckmann rearrangement? Give an example.	2
1	39.	Give an example for Beckmann rearrangement.	2
1	40.	Define Sommelet Hauser rearrangement? Give an example.	2
1	41.	What is Sommelet Hauser rearrangement? Give an example.	2
1	42.	Define Meisenheimer rearrangement and write its reaction.	2
1	43.	Give an example for Sommelet Hauser rearrangement.	2

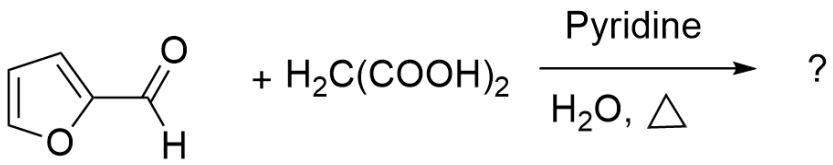
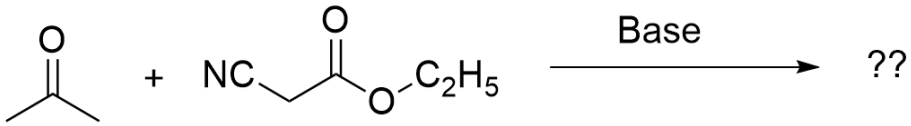
1	44.	Define Bayer-Villiger rearrangement and give the reaction.	2
1	45.	With an example, write Bayer-Villiger rearrangement reaction.	2
1	46.	What is Allylic rearrangement? Give an example.	2
1	47.	Mention the reagent used and the type of the product formed in the Bayer-Villeger and Allylic rearrangement.	2
1	48.	What is Neber rearrangement? Give an example.	2
1	49.	Give an example for Neber rearrangement.	2
1	50.	What is Japp-Klingemann rearrangement? Write its reaction	2
1	51.	Define Japp-Klingemann rearrangement and give the reaction.	2
1	52.	With example, write allylic rearrangement reaction.	2
1	53.	What is Neber rearrangement? Give reaction.	2
1	54.	Mention the reagent used in Neber and Japp-Klingemann rearrangement.	2
1	55.	Write a reaction of Neber rearrangement.	2
1	56.	Write an example for Meisenheimer rearrangement.	2
2	57.	Give any two applications of LDA.	2
2	58.	Comment on the uses of Gilmann reagent.	2
2	59.	List out the uses of DCC.	2
2	60.	Mention any two applications of DDQ.	2
2	61.	Comment on the uses of trialkyl silyl halides	2
2	62.	Give the applications of trimethyl silyl cyanide	2
2	63.	What is phase transfer catalyst? Give an example.	2
2	64.	Comment on the uses of phase transfer catalyst.	2
2	65.	What are crown ethers? How are they used in organic synthesis?	2
2	66.	What are crown ethers? Mention their applications.	2
2	67.	List out the uses of crown ethers.	2
2	68.	Write any two applications of Cyclodextrins.	2
2	69.	What are the specific uses of Fenton's reagent?	2

2	70.	What is Fenton's reagent? Give example.	2
2	71.	Write the uses of Fenton's reagent.	2
2	72.	Give any two applications of Ziegler-Natta catalyst.	2
2	73.	Comment on the uses of Ziegler-Natta catalyst.	2
2	74.	Write the resonance structures of diazomethane.	2
2	75.	Write the preparation and resonance forms of diazomethane.	2
2	76.	Give the uses of diazomethane.	2
2	77.	Give the applications of tributyl tinhydride.	2
2	78.	Comment on the uses of stannous chloride.	2
2	79.	Write a short note on Sharpless epoxide.	2
2	80.	Write the applications of Woodward.	2
2	81.	List out the applications of Prevost hydroxylation.	2
2	82.	Mention the uses of Stark enamine reaction.	2
2	83.	Give an account of uses of ultra sound in organic synthesis.	2
2	84.	Give any two advantages of microwave assisted organic synthesis.	2
3	85.	Mention the principles and technologies used in one group disconnection approach.	2
3	86.	Define Robinson Annulation Reaction and give one of its applications.	2
3	87.	Formulate any two synthon and their synthetic intermediate.	2
3	88.	Assess what happens when acetic acid is treated with hydrozoic acid in presence of concentrated sulphuric acid?	2
3	89.	write the reaction involved in Robinson annulations and mention one application	2
3	90.	How do you protect thiol group? Give example.	2
3	91.	Mention the characteristics of good protecting groups.	2
3	92.	Describe Perkin reaction and mention the catalyst used.	2
3	93.	Write the reaction involved in Claisen reaction.	2

3	94.	Suggest one reagent to protect thiol group with illustration.	2
3	95.	Suggest one reagent to protect Amine group with illustration.	2
3	96.	Suggest one reagent to protect carboxylic acid group with illustration.	2
3	97.	Mention the basic principles and technologies used in one group disconnection.	2
3	98.	What is one-group disconnection? Give an example.	2
3	99.	What are synthons and synthetic equivalents? Give example.	2
3	100.	Mention two applications of Darzens glysidic ester condensation.	2
3	101.	Define Keto-enol Tatomerism with examples.	2
3	102.	Give any two application of Aldol condensation.	2
3	103.	Define Darzens glysidic ester condensation with an example.	2
3	104.	Mention the criteria's for a good protecting group.	2
3	105.	What is Michael addition? Give an example.	2
3	106.	Write any two synthon and synthetic intermediates.	2
3	107.	Write the reaction involved in benzoin condensation.	2
3	108.	Give equations of the Perkin reactions.	2
3	109.	What is Tischenko Reaction? Give an example.	2
3	110.	Which enol form is more stable and why? 	2
3	111.	Give the enol form for the following ketones and identify the stable form. 	2
3	112.	Which Ketone will favour enol A or B, why?	2

			
3	113.	Predict the product and Name the following reaction . 	2
3	114.	Predict the product and Name the following reaction. 	2
3	115.	Predict the product and Name the following reaction. 	2
3	116.	Predict the products: 	2
3	117.	Predict the following reactions:  acetaldehyde malonic acid	2
3	118.	Predict the following reactions:  2-oxopropanoic acid	2
3	119.	Predict the following reaction:	2

		 acetic acid	

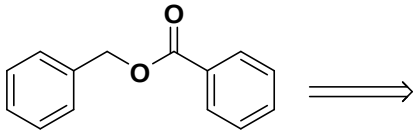
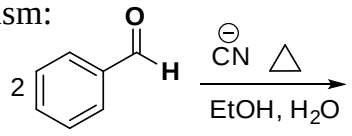
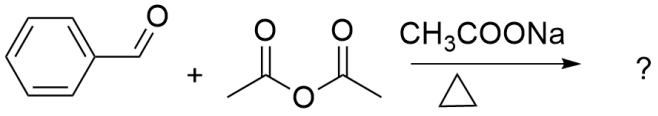
3	120.	Predict the following reaction:  furan-2-carbaldehyde	2
3	121.	Predict the following reaction:  	2
1	122.	Explain inter and intramolecular rearrangement.	4
1	123.	Explain the types molecular rearrangement.	4
1	124.	What is molecular rearrangement? Explain its types.	4
1	125.	Explain the mechanism of Pinacol-Pinacolone rearrangement.	4
1	126.	Discuss Pinacol-Pinacolone rearrangement with an example.	4
1	127.	Formulate the mechanism involved in Pinacol-Pinacolone rearrangement with an example.	4
1	128.	Explain the mechanism of Wagner-Meerwein rearrangement.	4
1	129.	Discuss the mechanism of Wagner-Meerwein rearrangement with an example.	4
1	130.	Discuss Benzidine rearrangement.	4
1	131.	Explain the mechanism of Benzidine rearrangement.	4
1	132.	Give the mechanism of Benzidine rearrangement.	4
1	133.	Discuss Demjanove rearrangement.	4
1	134.	Explain the mechanism of Demjanove rearrangement with an example.	4
1	135.	Give the mechanism of Demjanove rearrangement.	4

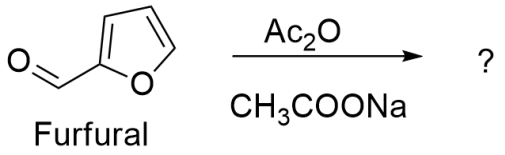
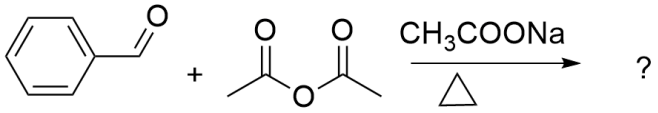
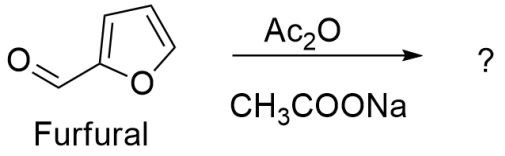
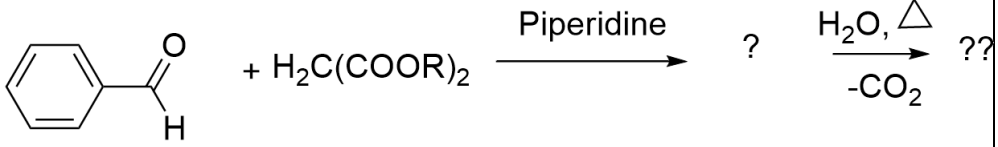
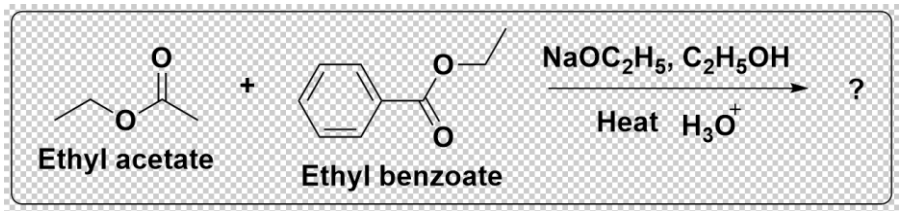
1	136.	Discuss Benzilic rearrangement.	4
1	137.	Explain the mechanism of Benzilic acid rearrangement with an example.	4
1	138.	Give the mechanism of Benzilic acid rearrangement.	4
1	139.	Explain the mechanism of Favorskii rearrangement with an example.	4
1	140.	Give the mechanism of Favorskii rearrangement.	4
1	141.	Discuss Arnt-Eistert synthesis.	4
1	142.	Explain the mechanism of Arnt-Eistert synthesis with an example.	4
1	143.	Give the mechanism of Arnt-Eistert synthesis.	4
1	144.	Discuss Fries rearrangement.	4
1	145.	Explain the mechanism of Fries rearrangement with an example.	4
1	146.	Give the mechanism of Fries rearrangement.	4
1	147.	Discuss Steven's rearrangement.	4
1	148.	Elaborate the mechanism of Steven's rearrangement with an example.	4
1	149.	Give the mechanism of Steven's rearrangement.	4
1	150.	Write a note on Dienophine rearrangement.	4
1	151.	Discuss the mechanism of Dienophine rearrangement.	4
1	152.	Write a note on Hofmann rearrangement.	4
1	153.	Explain the mechanism of Hofmann rearrangement with an example.	4
1	154.	Give the mechanism of Hofmann rearrangement.	4
1	155.	Discuss Curtius rearrangement.	4
1	156.	Discuss the mechanism of Curtius rearrangement with an example.	4
1	157.	Give the mechanism of Curtius rearrangement.	4
1	158.	Discuss Lossen rearrangement.	4

1	159.	Explain the mechanism of Lossen rearrangement with an example.	4
1	160.	Give the mechanism of Lossen rearrangement.	4
1	161.	Discuss Schmidt rearrangement.	4
1	162.	Explain the mechanism of Schmidt rearrangement with an example.	4
1	163.	Give the mechanism of Schmidt rearrangement.	4
1	164.	Discuss Sommelet Hauser rearrangement.	4
1	165.	Explain the mechanism of Sommelet Hauser rearrangement with an	4
1	166.	Give the mechanism of Sommelet Hauser rearrangement.	4
1	167.	Discuss Beckmann rearrangement.	4
1	168.	Explain the mechanism of Beckmann rearrangement with an example.	4
1	169.	Give the mechanism of Beckmann rearrangement.	4
1	170.	Discuss Bayer-Villiger rearrangement.	4
1	171.	Explain the mechanism of Bayer-Villiger rearrangement with an example.	4
1	172.	Give the mechanism of Bayer-Villiger rearrangement.	4
1	173.	Discuss Allylic rearrangement.	4
1	174.	Explain Allylic rearrangement with an example.	4
1	175.	Give the mechanism of Allylic rearrangement.	4
1	176.	Discuss Neber rearrangement.	4
1	177.	Elaborate the mechanism of Neber rearrangement with an example.	4
1	178.	Give the mechanism of Neber rearrangement.	4
1	179.	Explain Japp-Klingemann rearrangement.	4
1	180.	Discuss Japp-Klingemann rearrangement.	4
1	181.	Give the mechanism of Meisenheimer rearrangement.	4

1	182.	Explain Meisenheimer rearrangement.	4
2	183.	Give the applications of LDA.	4
2	184.	Write any two synthetic applications of Gilmann reagent and DCC.	4
2	185.	What is Gilmann reagent? Give its applications.	4
2	186.	Discuss on the synthetic applications of DCC.	4
2	187.	Discuss the applications of LDA and DCC in organic synthesis.	4
2	188.	Write the applications of DDQ.	4
2	189.	Comment on the applications of trialkyl silyl halides.	4
2	190.	Write any two synthetic applications of trialkyl silyl halides and trimethyl silyl cyanide	4
2	191.	Discuss the uses of trimethyl silyl cyanide	4

2	192.	With suitable example, write the synthetic applications of trimethyl silyl cyanide.	4
2	193.	Write a note on phase transfer catalyst.	4
2	194.	Write a brief account on phase transfer catalyst.	4
2	195.	What are crown ethers? Give its applications.	4
2	196.	Comment on the uses of Cyclodextrins.	4
2	197.	Write a note on Fenton's reagent.	4
2	198.	Using Ziegler-Natta as catalyst , write any two reactions.	4
2	199.	What is Ziegler-Natta catalyst? Explain with examples.	4
2	200.	Illustrate the synthetic applications of diazomethane.	4
2	201.	Illustrate with example, the use of tributyl tinhydride in organic synthesis.	4
2	202.	Comment on the uses of tributyl tinhydride.	4
2	203.	Elaborate the uses of stannous chloride.	4
2	204.	Write a note on the uses of Sharpless epoxide.	4
2	205.	Discuss the uses of Woodward reaction.	4
2	206.	Illustrate with example, the use of Prevost hydroxylation.	4
2	207.	Give the uses of Stark enamine reaction.	4
2	208.	Give a brief note on microwave induced organic synthesis.	4
2	209.	Explain microwave organic synthesis.	4
2	210.	Give an account of microwave induced organic synthesis.	4
2	211.	Write a note on ultra sound in organic synthesis.	4
2	212.	Write a note on polymer supported reagents.	4
2	213.	Write a short note on microwave induced organic synthesis.	4
3	214.	Write a note on Stobbe condensation	4

3	215.	Demonstrate Darzens glysidic ester condensation.	4
3	216.	Design any two protection and deprotection strategies for following groups i. Thiol ii. Carbonyl	4
3	217.	Give an account of Schmidt Reaction	4
3	218.	Give an account on the application of Aldol Condensation	4
3	219.	Formulate the retro-analysis of Nuciferol	4
3	220.	Explain Darzens glysidic ester condensation.	4
3	221.	Give the mechanism of aldol condensation of 1-butanal.	4
3	222.	Give all possible products formed by aldol condensation reaction between ethanal and propanal	4
3	223.	Give equations of the following name reactions: (i) Benzoin condensation (ii) Knoevenagel reaction (iii) Tischenko reaction (iv) Stobbe condensation	4
3	224.	What is Cannizaro's Condensation? Explain its mechanism.	4
3	225.	write a note on Storke enamine reaction.	4
3	226.	Design any two protection and deprotection strategies for following groups i. -OH ii. -SH	4
3	227.	Give synthon and synthetic equivalents obtained in the following retro analysis of 	4
3	228.	Predict the product with Mechanism: 	4
3	229.	Predict the product with Mechanism 	4

3	230.	Predict the product with Mechanism  Furfural	4
3	231.	Predict the product with Mechanism 	4
3	232.	Predict the product with Mechanism  Furfural	4
1	1.	Illustrate the mechanism of Pinacol-Pinacolone rearrangement.	5
1	2.	Formulate the mechanism involved in Pinacol-Pinacolone rearrangement.	5
1	3.	Outline the mechanism of Pinacol-Pinacolone rearrangement with an example.	5
1	4.	Predict the products with mechanism.  benzaldehyde	5
1	5.	Write products with mechanism:  Ethyl acetate Ethyl benzoate	5
1	6.	Discuss the mechanism involved in Pinacol-Pinacolone rearrangement.	5

1	7.	Write a note on Wagner-Meerwein rearrangement.	5
1	8.	Explain the mechanism of Wagner-Meerwein rearrangement.	5
1	9.	Discuss the mechanism of Benzidine rearrangement with an example.	5
1	10.	Formulate the mechanism of Benzidine rearrangement.	5
1	11.	Illustrate the mechanism of Demjanove rearrangement.	5
1	12.	Discuss the mechanism of Demjanove rearrangement with an example.	5
1	13.	Outline the mechanism of Demjanove rearrangement.	5
1	14.	Illustrate the mechanism of Benzilic acid rearrangement.	5
1	15.	Discuss the mechanism of Benzilic acid rearrangement with an example.	5
1	16.	Formulate the mechanism of Benzilic acid rearrangement.	5
1	17.	Write a note on Favorskii rearrangement.	5
1	18.	Explain Favorskii rearrangement.	5
1	19.	Outline the mechanism of Favorskii rearrangement.	5
1	20.	Discuss Arnt-Eistert synthesis.	5
1	21.	Explain the mechanism of Arnt-Eistert synthesis with an example.	5
1	22.	Give the mechanism of Arnt-Eistert synthesis.	5
1	23.	Formulate the mechanism of Fries rearrangement.	5
1	24.	Discuss the mechanism of Fries rearrangement with an example.	5
1	25.	Outline the mechanism of Fries rearrangement.	5
1	26.	Illustrate the mechanism of Steven's rearrangement.	5
1	27.	Discuss the mechanism of Steven's rearrangement with an example.	5
1	28.	Outline the mechanism of Steven's rearrangement.	5

1	29.	Elaborate Dienophine rearrangement	5
1	30.	Explain Dienophine rearrangement	5
1	31.	Illustrate the mechanism of Hofmann rearrangement.	5
1	32.	Write a note on Hofmann rearrangement.	5
1	33.	Outline the mechanism of Hofmann rearrangement.	5
1	34.	Illustrate the mechanism of Curtius rearrangement.	5
1	35.	Discuss the mechanism of Curtius rearrangement with an example.	5
1	36.	Outline the mechanism of Curtius rearrangement.	5
1	37.	Illustrate the mechanism of Lossen rearrangement.	5
1	38.	Discuss the mechanism of Lossen rearrangement with an example.	5
1	39.	Outline the mechanism of Lossen rearrangement.	5
1	40.	Illustrate the mechanism of Schmidt rearrangement.	5
1	41.	Discuss the mechanism of Schmidt rearrangement with an example.	5
1	42.	Outline the mechanism of Schmidt rearrangement.	5
1	43.	Illustrate the mechanism of Beckmann rearrangement.	5
1	44.	Formulate the mechanism of Beckmann rearrangement with an example.	5
1	45.	Outline the mechanism of Beckmann rearrangement.	5
1	46.	Illustrate the mechanism of Sommelet Hauser rearrangement.	5
1	47.	Discuss the mechanism of Sommelet Hauser rearrangement with an example.	5
1	48.	With mechanism explain Sommelet Hauser rearrangement.	5
1	49.	Illustrate the mechanism of Bayer-Villiger rearrangement.	5
1	50.	Discuss the mechanism of Bayer-Villiger rearrangement with an example.	5

1	51.	What is Bayer-Villiger rearrangement? Explain the mechanism involved in it	5
1	52.	Illustrate the mechanism of Allylic rearrangement.	5
1	53.	Discuss the mechanism of Allylic rearrangement with an example.	5
1	54.	Outline the mechanism of Allylic rearrangement.	5
1	55.	Explain the mechanism of Neber rearrangement.	5
1	56.	Elaborate the mechanism of Neber rearrangement with an example.	5
1	57.	Formulate the mechanism of Japp-Klingemann rearrangement	5
1	58.	Write a note on Japp-Klingemann rearrangement	5
1	59.	Explain Japp-Klingemann rearrangement	5
1	60.	Write the mechanism of Meisenheimer rearrangement.	5
1	61.	Give the mechanism of Meisenheimer rearrangement with an example.	5
1	62.	Give the mechanism of Meisenheimer rearrangement.	5
1	63.	Write a note on Neber rearrangement.	5
2	64.	Comment on the uses of LDA taking suitable examples.	5
2	65.	Give the applications of LDA and DCC in organic synthesis.	5
2	66.	What is Gilmann reagent? Write the applications.	5
2	67.	Explain the role of DCC in organic synthesis.	5
2	68.	Write a note on the uses of DDQ.	5
2	69.	Discuss the uses of trialkyl silyl halides	5
2	70.	Write the synthetic applications of trimethyl silyl cyanide with suitable example.	5
2	71.	Elaborate the synthetic applications of trimethyl silyl cyanide	5

2	72.	How tetra butyl ammonium bromide acts as a phase transfer catalyst? Write the mechanism.	5
2	73.	Write a brief account on phase transfer catalyst.	5
2	74.	Comment on the uses of phase transfer catalyst.	5
2	75.	Write a note on crown ethers.	5
2	76.	What are crown ethers? Give its applications.	5
2	77.	Discuss the uses of crown ethers.	5
2	78.	Give the uses of Cyclodextrins.	5
2	79.	Comment on the uses of Fenton's reagent.	5
2	80.	What is Fenton's reagent? Elaborate its applications.	5
2	81.	What is Ziegler-Natta catalyst? Explain with examples.	5
2	82.	Using Ziegler-Natta as catalyst, give any two reactions.	5
2	83.	Discuss the applications of Ziegler-Natta catalyst.	5
2	84.	With examples write a note on the uses of diazomethane.	5
2	85.	Illustrate with example, the applications of tributyl tinhydride in organic synthesis.	5
2	86.	Write the uses of tributyl tinhydride.	5
2	87.	Give an account on the applications of stannous chloride.	5
2	88.	What are Sharpless epoxides? Elaborate its applications.	5
2	89.	Write a note on Woodward and Prevost hydroxylation.	5
2	90.	Comment on the uses of Woodward.	5
2	91.	Illustrate with example, the use of Prevost hydroxylation.	5
2	92.	Comment on the uses of Prevost hydroxylation.	5
2	93.	Elaborate the uses of Stark enamine reaction.	5

2	94.	Give an account of microwave induced organic synthesis.	5
2	95.	Illustrate the advantages of microwave synthesis over conventional organic synthesis. Mention its application	5
2	96.	Write an account on application of polymeric reagents in organic synthesis.	5
2	97.	Discuss ultra sound in organic synthesis.	5
3	98.	Give an account on the application of Aldol Condensation	5
3	99.	Predict the retrosynthetic approach of Saccharine.	5
3	100.	Write a note on Stobbe Condensation.	5
3	101.	What is Claisen reaction? Write its application.	5
3	102.	Write a note on Cannizzaro's Reaction	5
3	103.	what are the functional groups present in Penicillin- V ? Discuss its retrosynthesis.	5
3	104.	Illustrate Schmidt reaction.	5
3	105.	Write the different protecting group used to protect amino group. How are they deprotected? Illustrate.	
3	106.	How protection and deprotection of carboxylic group is carried out in an organic reaction.	5
3	107.	Illustrate Knoevengel reaction.	5
3	108.	Outline the retrosynthesis and synthesis of benzofuran.	5
3	109.	Outline the retrosynthesis and synthesis of Saccharine.	5
3	110.	Outline the retrosynthesis and synthesis of α-bisabolene.	5
3	111.	What is Michael addition? Write its application.	5
3	112.	Illustrate one group disconnection approach with suitable examples.	5
3	113.	Outline the retrosynthesis and synthesis of p-methoxy acetophenone.	5
	114.		

1	1.	What is Pinacol-Pinacolone rearrangement? Explain its mechanism.	6
1	2.	What is Pinacol-Pinacolone rearrangement? Discuss the mechanism with an example?	6
1	3.	Write a note on Pinacol-Pinacolone rearrangement.	6
1	4.	Explain the reaction and mechanism of Wagner-Meerwin and Benzidine rearrangement.	6
1	5.	Write a note on Benzidine rearrangement.	6
1	6.	What is Benzidine rearrangement? Outline its mechanism.	6
1	7.	What is Benzidine rearrangement? Discuss the mechanism with an example.	6
1	8.	Write a note on Demjanove rearrangement.	6
1	9.	What is Demjanove rearrangement? Outline its mechanism.	6
1	10.	Discuss the mechanism of Demjanove rearrangement with an example.	6
1	11.	Write a note on Benzilic acid rearrangement.	6
1	12.	What is Benzilic acid rearrangement? Outline its mechanism.	6
1	13.	Discuss the mechanism of Benzilic acid rearrangement with an example.	6
1	14.	With suitable example, explain Benzilic acid rearrangement.	6
1	15.	Give the mechanism of Favorskii rearrangement and give any two evidences in favour of the mechanism.	6
1	16.	Write a note on Favorskii rearrangement.	6
1	17.	What is Favorskii rearrangement? Outline its mechanism.	6
1	18.	Discuss the mechanism of Favorskii rearrangement with an example.	6
1	19.	Give an example for Arnt-Eistert synthesis. Write the mechanism of the rearrangement involved in the synthesis.	6

1	20.	Write a note on Arnt-Eistert synthesis.	6
1	21.	Outline the mechanism Arnt-Eistert synthesis.	6
1	22.	Discuss the mechanism of Arnt-Eistert synthesis with an example.	6
1	23.	Write a note on Fries rearrangement.	6
1	24.	What is Fries rearrangement? Outline its mechanism.	6
1	25.	Discuss the mechanism of Fries rearrangement with an example.	6
1	26.	Write a note on Steven's rearrangement.	6
1	27.	What is Steven's rearrangement? Outline its mechanism.	6
1	28.	Discuss the mechanism of Steven's rearrangement with an example.	6
1	29.	Write a note on Dienophine rearrangement rearrangement.	6
1	30.	What is Dienophine rearrangement? Outline its mechanism.	6
1	31.	Discuss the mechanism of Dienophine rearrangement with an example.	6
1	32.	Write a note on Hofmann rearrangement.	6
1	33.	What is Hofmann rearrangement? Outline its mechanism.	6
1	34.	Discuss the mechanism of Hofmann rearrangement with an example.	6
1	35.	Explain and sketch the mechanism of Curtius and Hofmann rearrangement	6
1	36.	Write a note on Curtius rearrangement.	6
1	37.	What is Curtius rearrangement? Outline its mechanism.	6
1	38.	Discuss the mechanism of Curtius rearrangement with an example.	6
1	39.	Write a note on Lossen rearrangement.	6
1	40.	What is Lossen rearrangement? Outline its mechanism.	6
1	41.	Discuss the mechanism of Lossen rearrangement with an example.	6

1	42.	Explain the reaction mechanism of Lossen and Schmidt rearrangement.	6
1	43.	Write a note on Schmidt rearrangement.	6
1	44.	What is Schmidt rearrangement? Outline its mechanism.	6
1	45.	Discuss the mechanism of Schmidt rearrangement with an example.	6
1	46.	Write a note on Beckmann rearrangement.	6
1	47.	What is Beckmann rearrangement? Outline its mechanism.	6
1	48.	Discuss the mechanism of Beckmann rearrangement with an example.	6
1	49.	Write a note on Sommelet Hauser rearrangement.	6
1	50.	What is Sommelet Hauser rearrangement? Outline its mechanism.	6
1	51.	Discuss the mechanism of Sommelet Hauser rearrangement with an example.	6
1	52.	Write a note on Bayer-Villiger rearrangement.	6
1	53.	What is Bayer-Villiger rearrangement? Outline its mechanism.	6
1	54.	Discuss the mechanism of Bayer-Villiger rearrangement with an example.	6
1	55.	What is Beckmann rearrangement? Outline its mechanism.	6
1	56.	Discuss the mechanism of Beckmann rearrangement with an example.	6
1	57.	Write a note on Allylic rearrangement.	6
1	58.	What is Allylic rearrangement? Outline its mechanism.	6
1	59.	Discuss the mechanism of Allylic rearrangement with an example.	6
1	60.	Write a note on Neber rearrangement.	6
1	61.	Write a note on Japp-Klingemann rearrangement and Meisenheimer rearrangements	6
1	62.	Discuss Japp-Klingemann rearrangement and Meisenheimer rearrangements	6

1	63.	What is Japp-Klingemann rearrangement? Explain its mechanism.	6
1	64.	Define Meisenheimer rearrangement. Elaborate its mechanism.	6
1	65.	Explain Japp-Klingemann rearrangement and Meisenheimer rearrangements	6
1	66.	What is Neber rearrangement? Outline its mechanism.	6
2	67.	Comment on the uses of LDA.	6
2	68.	Discuss the use of the following reagents in organic synthesis. i) Gilmann reagent ii) DCC	6
2	69.	Comment on the uses of Gilmann reagent and LDA.	6
2	70.	Discuss the applications of DCC and LDA	6
2	71.	Give the applications of DDQ and trimethyl silyl cyanide.	6
2	72.	Comment on the uses of DDQ and trialkyl silyl halides.	6
2	73.	What are the applications of trialkyl silyl halides and trimethyl silyl cyanide?	6
2	74.	Discuss the uses of cyclodextrin and crown ethers.	6
2	75.	Elaborate the synthetic applications of phase transfer catalyst and Cyclodextrins.	6
2	76.	What is phase transfer catalyst? Elaborate its uses.	6
2	77.	Write a note on the uses of phase transfer catalyst and crown ethers.	6
2	78.	What are crown ethers? Elaborate its uses.	6
2	79.	Elaborate the applications of cyclodextrins and crown ethers.	6
2	80.	Discuss the applications of the following i) Fenton's reagent ii) Ziegler-Natta catalyst	6
2	81.	Explain the applications of the following i) Fenton's reagent ii) Ziegler-Natta catalyst	6
2	82.	Comment on the uses of Fenton's reagent and Ziegler-Natta catalyst.	6

2	83.	What is Ziegler-Natta catalyst? Elaborate its applications.	6
2	84.	Comment on the uses of diazomethane and tributyl tinhydride.	6
2	85.	Outline the uses of tributyl tinhydride and stannous chloride.	6
2	86.	Discuss the uses of stannous chloride and sharpless epoxide.	6
2	87.	Outline the uses of tributyl tinhydride and sharpless epoxide.	6
2	88.	Comment on the uses of sharpless epoxide and stannous chloride.	6
2	89.	Write a note on Woodward and Prevost hydroxylation.	6
2	90.	Write the uses of Woodward and Stark enamine reaction .	6
2	91.	Write a note on Stark enamine reaction and Prevost hydroxylation.	6
2	92.	Give an account of microwave induced organic synthesis.	6

2	93.	Illustrate the advantages of microwave synthesis over conventional organic synthesis. Mention its application	6
2	94.	Discuss polymer supported reagents and their synthesis	6
2	95.	Write a note on polymer supported reagents and their synthesis	6
3	96.	Illustrate on i. Tischenko Reaction ii. Knoevenagel Reaction	6
3	97.	Predict the retrosynthesis and synthesis approach of α-bisbolene and saccharine	6
3	98.	Explain the Perkin and Claisen reaction and their mechanism	6
3	99.	Demonstrate the reaction and mechanism of Benzoin and Stobbe	6
3	100.	Outline the synthesis of peptide by protecting amino group using different protecting agents	6
3	101.	Suggest retrosynthetic analysis of penicillin-V. Provide one forward synthesis with intermediates and reagents	6
3	102.	Predict the retrosynthesis and synthesis approach of benzofurans and p-methoxy acetophenone	6
3	103.	Write notes on i. Cannizaros reaction ii Michael addition	6
3	104.	Outline the protection and deprotection action of hydroxyl and thiol groups taking suitable examples	6
3	105.	Demonstrate the reaction and mechanism of Knoevenagel and Tischenko reaction.	6
3	106.	Predict the retrosynthesis and complete synthesis approach of	6
3	107.	Illustrate on 1. Darzens glysidic ester condensation 2. Michael addition.	6
3	108.	Compare one group disconnection and two group disconnection using suitable example.	6

****

Question Paper Pattern- Model Question Paper

Q.P Code: 55103

St. Philomena's College (Autonomous) Mysore
II Semester M.Sc. - Final Examination : May - 2019
Subject: ORGANIC CHEMISTRY
Title: Synthetic Organic Chemistry and Molecular Rearrangements

Time: 3 Hours **Max Marks: 70**

PART –A

Answer the following: **10×2=20**

a. Point out the use of NBS in organic synthesis.

b. Write the reaction involved in the Claisen reaction.

c. What are the specific uses of Fenton's reagent?

d. Give an account of uses of ultra sound in organic synthesis.

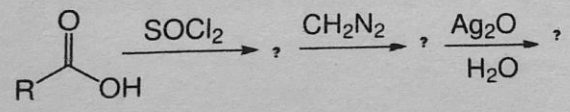
e. Name the reagent used and also the type of the product formed in the Curtius and Schmidt rearrangement.

f. Illustrate the role of Lossen rearrangement.

g. Suggest any one reagent to protect thiol group with illustration.

h. Mention the basic principles and technologies used in one group disconnection approach.

i. Predict the products in the following reaction:



j. Define Robinson annulations reaction and give one of its applications.

PART –B

Answer any FIVE of the following: **5×10=50**

2. a. Give an account of microwave induced organic synthesis.

b. How tetra butyl ammonium bromide acts as a phase transfer catalyst? Write the mechanism. **5+5**

PTO

3. a. Outline the synthesis of peptide by protecting amino group using different protecting agents.

   b. Using Ziegler – Natta as a catalyst write any two reactions. **6+4**

4. a. Discuss the mechanism involved in Pinacol-pinacolone rearrangement.

   b. Write a note on crown ethers. **5+5**

5. a. Formulate the mechanism involved in Beckmann rearrangement.

   b. Predict the retrosynthesis and synthesis approach of benzofuran. **4+6**

6. a. Comment on the uses of LDA taking suitable examples.

   b. Write a note on Stobbe condensation. **5+5**

7. a. Give an account on the applications of Aldol condensation.

   b. Outline the protection action of hydroxyl group taking sutiable examples. **5+5**

8. a. What is Bayer – Villeger rearrangement? Explain the mechanism involved in it.

   b. Write notes on: **4+3+3**

      i) Michael addition reaction

      ii) Tischenko reaction

****